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Image processing device, image processing method, and non-transitory computer readable medium

Abstract

An image processing apparatus includes an image obtainer that obtains a moving image including an operation area captured with an image sensor, an image generator that generates an image of a face region of a worker performing an operation in the operation area based on the moving image, a statistical information generator that generates statistical information for the operation performed by the worker based on the moving image, a data generator that generates data including the generated image of the face of the worker linked to the generated statistical information for the operation performed by the worker, and an output unit that outputs the generated data.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a technique for analyzing the behavior of a target using image sensors.

BACKGROUND

(2) The market of factory automation provides practical techniques for facilitating, for example, an increase in productivity by identifying, for example, the number of workers performing operations and machines in operation and their operation positions using an image sensor installed on a cellular line and analyzing the details of operations in each operational process.

(3) For example, Patent Literature 1 describes a technique for measuring the operation time by reading an identification (ID) tag attached to a workpiece operating on a line at an entrance and an exit of the line and identifying the operation start time and the operation end time for the workpiece. Patent Literature 2 describes a technique for displaying a time chart or a moving image associated with the details of operations performed by a tracked worker over a series of operations in an operational process.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2006-202255

Patent Literature 2: Japanese Unexamined Patent Application Publication No. 2018-073176

SUMMARY

Technical Problem

(5) However, known techniques involve operations for specifying targets to be analyzed for the details of operations, such as attaching ID tags to workpieces or specifying a person whose operations are to be tracked by a user. For a technique using ID tags, the detection accuracy of a sensor or another device for reading the ID tags affects the accuracy of the measured time. The ID tags are also to be read for every process. For a user to specify a worker to be tracked in a video including a line, the user cannot distinguish individual workers. Thus, the user can identify the details of operations performed by each worker but without identifying individual workers.

(6) In response to the above issue, one or more aspects of the present invention are directed to a technique for analyzing the details of an operation performed and also identifying an individual worker performing the operation based on an image obtained with an image sensor.

Solution to Problem

(7) The device, method, and program according to one or more aspects of the present invention have the structure below.

(8) An image processing apparatus according to a first aspect of the present invention includes an image obtainer that obtains a moving image including an operation area captured with an image sensor, an image generator that generates an image of a face region of a worker performing an operation in the operation area based on the moving image, a statistical information generator that generates statistical information for the operation performed by the worker based on the moving image, a data generator that generates data including the generated image of the face of the worker linked to the generated statistical information for the operation performed by the worker, and an output unit that outputs the generated data. This structure links an individual worker and the statistical information with each other based on the moving image from the image sensor, and allows analysis of operations identifying the individual worker.

(9) The data generator may generate data including images of face regions of a plurality of workers generated by the image generator linked to statistical information for operations performed by the plurality of workers generated by the statistical information generator. The data generator may generate, for data output from the output unit, new data including at least two images of the images of the face regions of the plurality of workers integrated with the statistical information linked to the at least two images. The at least two images may be selected based on a user input. This allows

the multiple images generated for the face region of a single worker to be integrated into a single image, and allows integration of the statistical information linked to the multiple images.

(10) The data generator may generate, for data output from the output unit, new data including the image of the face region of the worker replaced by another image of the face region of the worker generated by the image generator. The data generator may generate data including an image of the face region of the worker selected based on a criterion including at least one of a size, an orientation, or a facial feature of the face region of the worker linked to the generated statistical information for the operation performed by the worker. The data generator may generate, based on matching between images of face regions of a plurality of workers obtained in advance and the generated image of the face region of the worker, data including an image of the face region of the worker selected from the images obtained in advance linked to the generated statistical information for the operation performed by the worker. This allows an image of the face region of a worker to be changed to an image more identifiable as the worker, and facilitates identification of the statistical information for the worker.

(11) The statistical information generator may generate outline information and detailed information about the statistical information for the operation performed by the worker based on the moving image. The data generator may generate data including the image of the face region of the worker linked to the outline information about the generated statistical information for the operation performed by the worker and the detailed information about the generated statistical information for the operation performed by the worker. Linking the image of the face of the worker to various items of information useful for analyzing the operations of the worker allows more efficient analysis of operations.

(12) The data generator may generate a moving image associated with the generated statistical information for the operation performed by the worker based on the obtained moving image. The data generator may generate data including the image of the face region of the worker linked to the generated statistical information for the operation performed by the worker and the generated moving image. The statistical information generator may generate, based on the moving image, statistical information for the operation performed by the worker for each of one or more operational processes included in the operation area. The data generator may generate, based on the obtained moving image, a moving image associated with the generated statistical information for the operation performed by the worker for each of the one or more operational processes. The data generator may generate data including the image of the face region of the worker linked to the generated statistical information for the operation performed by the worker for each of the one or more operational processes and the generated moving image for each of the one or more operational processes. This allows the moving image as a basis of the statistical information for the worker to be viewable, and allows more efficient analysis of operations.

(13) The statistical information generator may generate evaluation information indicating an evaluation result obtained based on a predetermined criterion for the statistical information for the operation performed by the worker generated based on the moving image. The data generator may generate data including the image of the face region of the worker linked to the generated statistical information for the operation performed by the worker and the evaluation information. This allows information usable in analyzing the statistical information, such as statistical information for the worker indicating the operation time largely above or below target values, to appear as the evaluation results, and allows more efficient analysis of operations.

(14) Other aspects of the present invention may be directed to an image processing method including at least part of the above processes, a program for causing a computer to implement the method, or a non-transitory computer-readable storage medium storing the program. The present invention may be implemented by combining the above structures and processes in any possible manner unless any technical contradiction arises.

Advantageous Effects

(15) The technique according to the above aspects of the present invention allows the details of operations to be analyzed and also an individual worker to be identified based on images obtained with an image sensor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an image processing apparatus according to one or more embodiments of the present invention showing an example structure.
- (2) FIG. 2 is a block diagram of a personal computer (PC) (image processing apparatus) according to an embodiment showing an example structure.
- (3) FIG. 3 is a flowchart of an example process performed by the PC according to the embodiment.
- (4) FIG. 4 is a schematic diagram of an example display screen in the embodiment.
- (5) FIG. 5 is a schematic diagram of an example display screen in a first modification.
- (6) FIG. 6 is a schematic diagram of an example display screen in a second modification.
- (7) FIG. 7 is a schematic diagram of an example display screen in a third modification.
- (8) FIG. 8 is a schematic diagram of an example display screen in a fourth modification.
- (9) FIG. 9 is a schematic diagram of an example display screen in a fifth modification.
- (10) FIG. 10 is a schematic diagram of an example display screen in a sixth modification.
- (11) FIG. 11 is a schematic diagram of an example display screen in a seventh modification.
- (12) FIG. 12 is a schematic diagram of an example display screen in an eighth modification.

DETAILED DESCRIPTION

(13) Example Use

(14) An example use of a technique according to one or more embodiments of the present invention will be described. To analyze the details of operations performed by each worker based on an image including the workers obtained with an image sensor located on a cellular or another line in a factory, the workers are to be individually identified. Identifying a worker includes positioning of the worker using the intensity of radio waves at a portable terminal carried by the worker and at wireless local area network (LAN) access points or positioning of the worker by reading an electronic tag attached to the worker. However, such positioning may have lower positioning accuracy with radio waves from the wireless LAN affected by various facilities and devices, or may involve preparation of many electronic tags to be attached to workers.

(15) FIG. 1 is a block diagram of an image processing apparatus **100** according to an embodiment of the present invention showing an example structure. The image processing apparatus **100** includes an image obtainer **101**, a face image generator **102**, a statistical information generator **103**, a data generator **104**, and an output unit **105**. The image obtainer **101** obtains an image captured with an image sensor such as a 360-degree camera located on a cellular line or another line. The face image generator **102** generates an image of a face region of each worker (face image) based on the captured image. The statistical information generator **103** generates statistical information for operations performed by each worker based on the captured image. The data generator **104** generates data including images of face regions of workers generated by the face image generator **102** linked to statistical information for operations performed by the workers generated by the statistical information generator **103**. The output unit **105** outputs the data generated by the data generator **104**.

(16) The image processing apparatus **100** according to one or more embodiments of the present invention can accurately identify workers based on a captured image captured with the image sensor and provide information indicating the details of analysis of operations performed by each worker.

(17) First Embodiment

(18) A first embodiment of the present invention will now be described.

(19) FIG. 2 is a schematic block diagram of an image processing system according to the first embodiment showing an example structure. The image processing system according to the first embodiment includes an image sensor, a personal computer (PC) **200** (image processing apparatus), and a display. The image sensor and the PC **200** are connected to each other with a wire or wirelessly. The PC **200** and the display are connected to each other with a wire or wirelessly.

(20) In the first embodiment, the image sensor captures a cellular line placed at a factory, a distribution warehouse, or another facility. An area of the cellular line in a captured image captured with the image sensor is a detection area for human detection targeting workers. The image sensor captures an image of the cellular line and outputs the captured image to the PC **200**. Although the image sensor may capture an image in any direction, the image sensor in the first embodiment is installed above the cellular line, or specifically, on the ceiling of the floor on which the cellular line is placed to provide a wide overall view of the cellular line. The PC **200** performs human detection based on the captured image captured with the image sensor and outputs detection results to the display. Examples of the display include a display device and an information processing terminal (e.g., a smartphone).

(21) In the first embodiment, the PC **200** is a device separate from the image sensor and the display. In some embodiments, the PC **200** may be integral with the image sensor or the display. The PC **200** may be at any location. The PC **200** may be in, for example, the same room as the image sensor. In some embodiments, the PC **200** may be a cloud computer.

(22) The PC **200** includes an input unit **210**, a controller **220**, a storage **230**, and an output unit **240**. The controller **220** includes an image generator **221**, a statistical information generator **222**, and a data generator **223**. The input unit **210**, the image generator **221**, the statistical information generator **222**, the data generator **223**, and the output unit **240** respectively correspond to the image obtainer, the image generator, the statistical information generator, the data generator, and the output unit in one or more embodiments of the present invention.

(23) The input unit **210** obtains, from the image sensor, moving image data including the cellular line captured with the image sensor, and outputs the moving image data to the controller **220**. The moving image data obtained by the input unit **210** from the image sensor is stored into the storage **230** by the controller **220**. The moving image data obtained by the input unit **210** includes the cellular line captured over a predetermined period (an hour, a day, or another period set as appropriate).

(24) The controller **220** includes, for example, a central processing unit (CPU), a random-access memory (RAM), and a read-only memory (ROM). The controller **220** controls each unit in the PC **200** and performs various information processes.

(25) The image generator **221** detects a face region of a worker in the captured image based on the moving image data obtained by the input unit **210**. A face region of a worker in moving image data obtained with the image sensor can be detected with known techniques by detecting the facial features of a human body. This process is not described in detail herein. The image generator **221** also determines whether a worker with a detected face region is a worker to be analyzed for the details of operations. In the present embodiment, the storage **230** prestores worker identifications (IDs) for identifying workers and information for identifying a worker with a detected face region as a worker to be analyzed based on, for example, a detected position in the moving image and the degree of similarity in image features. The image generator **221** identifies, based on the information stored in the storage **230**, the worker to be analyzed using the detected face region, and stores the worker ID of the identified worker and detected information about the position of the worker in the moving image into the storage **230**. When no worker is identified using the detected face region, the image generator **221** determines that the worker corresponding to the detected face region is a new worker, generates a new worker ID, and stores the generated worker ID and the detected position information in the moving image into the storage **230**.

(26) The image generator **221** generates an image of the detected face region. The image of the face region of the worker generated by the image generator **221** may be stored into the storage **230**, or may be transmitted to each unit in the PC **200** that processes the image.

(27) The statistical information generator **222** generates statistical information for the details of operations performed by each of the detected workers based on the moving image data obtained by the input unit **210** and the above information generated by the image generator **221**, which are stored in the storage **230**. An example of the statistical information includes a Gantt chart showing the operation time for each process on a cellular line. The statistical information may include information about a target operation time as a reference operation time for each process to be compared with the operation time of each worker. The operation time of each worker for each process can be calculated based on a captured image captured with the image sensor, and a Gantt chart can be generated based on the calculation results with known techniques. These processes are not described in detail herein.

(28) The data generator **223** generates data including the image of the face region of each worker identified by the image generator **221** linked to the statistical information for the details of operations performed by each worker generated by the statistical information generator **222**. In the present embodiment, the data generator **223** generates data such as an image displayable on the display connected to the output unit **240**. However, the data generator **223** may generate other data processable by other information processing devices.

(29) In addition to the above statistical information, the storage **230** stores a program executable by the controller **220** and various sets of data used by the controller **220**. For example, the storage **230** is an auxiliary storage device such as a hard disk drive or a solid state drive. The output unit **240** outputs an image generated by the data generator **223** to the display. The image generated by the data generator **223** may be stored into the storage **230** for output through the output unit **240** at any time.

(30) FIG. **3** is a flowchart of an example process performed by the PC **200**. The PC **200** performs the procedure in FIG. **3** after the input unit **210** obtains moving image data from the image sensor and stores the moving image data into the storage **230**.

(31) In step **S301**, the controller **220** first obtains the moving image data captured with the image sensor and stored in the storage **230**.

(32) The image generator **221** then repeats the processing in steps **S302** to **S305** for each frame of the moving image data obtained from the storage **230**. In step **S302**, the image generator **221** detects a face region of a worker from a frame of the moving image data obtained from the storage **230**. In step **S303**, the image generator **221** obtains, from the storage **230**, information for determining whether the worker is a worker to be analyzed based on the detected face region and identifies the worker corresponding to the face region detected in step **S302** as a worker to be analyzed. When a worker to be analyzed is successfully identified based on the detected face region, the image generator **221** stores the worker ID of the worker linked to information about the position of the worker in the frame into the storage **230**. When a worker to be analyzed is not identified based on the detected face region, the image generator **221** generates a new worker ID to the worker corresponding to the detected face region, and stores the generated worker ID linked to information about the position of the worker in the frame into the storage **230**.

(33) In step **S304**, the image generator **221** further distinguishes operational processes performed by the worker based on the information about the cellular line captured with the image sensor and stored in the storage **230** and the information about the position of the worker in the frame, and stores the distinguished operational process linked to the above information about the position of the worker into the storage **230**.

(34) In step **S305**, the image generator **221** generates an image of the face region of the worker detected in step **S302** based on the image in the frame, and stores the generated image of the face region linked to the worker ID into the storage **230**. In the present embodiment, the image

generator **221** obtains an image of a face region of a worker for each frame obtained by the input unit **210**. However, the image generator **221** repeating the processing in steps **S302** to **S305** may obtain an image of a face region of a worker intermittently at any time, or may simply obtain an image of a face region of each worker once.

(35) In step **S306**, the image generator **221** determines whether all frames of the moving image data obtained in step **S301** have undergone the processing in steps **S302** to **S305**. For a frame yet to undergo the processing in steps **S302** to **S305** (No in **S306**), the image generator **221** returns to step **S302** and performs the processing in steps **S302** to **S305** on the frame. When all frames of the moving image data have undergone the processing in steps **S302** to **S305** (Yes in **S306**), the image generator **221** advances the processing to step **S307**.

(36) In step **S307**, the statistical information generator **222** generates statistical information, such as the operation time for each operational process on the cellular line and the average operation time, based on the data for each worker stored in the storage **230** through the processing in steps **S302** to **S305** described above. In the present embodiment, the statistical information generated by the statistical information generator **222** includes Gantt charts indicating the operation time for each process. The statistical information generator **222** links generated Gantt charts for each worker to the corresponding worker IDs and stores the charts with the IDs into the storage **230**.

(37) In step **S308**, the data generator **223** obtains, from the storage **230**, the image of the face region of each worker generated by the image generator **221** and the statistical information generated by the statistical information generator **222**, and generates data including the image of the face region of each worker linked to the statistical information based on the worker ID.

(38) In step **S309**, the output unit **240** outputs the data generated by the data generator **223** in step **S308** to the display. In step **S309**, instead of or in addition to outputting the data generated by the data generator **223**, the output unit **240** may store the data into the storage **230**.

(39) FIG. **4** is a schematic diagram of an example display screen **400** on the display with data generated by the data generator **223** through the procedure in FIG. **3**. The display screen **400** shows face images **401** to **405** corresponding to the respective workers obtained by the image generator **221**, identification numbers (ID1 to ID5 in the figure) for the respective workers, and Gantt charts indicating the details of operations performed by the respective workers generated by the statistical information generator **222**.

(40) The display screen **400** shows the face images, the identification numbers, and the Gantt charts corresponding to the respective workers in a manner associated with one another for a user viewing the display screen **400**. In the present embodiment, the cellular line captured with the image sensor includes separate operational processes for a worker, or specifically, processes **1** to **5**, which are displayed in the Gantt charts **406** to **410** generated by the statistical information generator **222** indicating the operation time for each process. The display screen **400** also shows a Gantt chart **411** for target values generated by the statistical information generator **222** based on a target operation time preset for each process.

(41) The structure according to the present embodiment can thus accurately identify a worker and can link the worker to statistical information for the details of operations performed by the worker by generating data including a face image for each worker performing operations on a cellular line linked to statistical information for the operations based on moving image data from the image sensor capturing the cellular line, without causing the cost of, for example, attaching tags or other devices to workers or without any possibility of, for example, lower accuracy in identifying workers when a communication system such as a wireless LAN is used to identify workers.

(42) Others

(43) The above embodiment describes exemplary structures according to one or more aspects of the present invention. The present invention is not limited to the specific embodiment described above, but may be modified variously within the scope of the technical ideas of the invention. For example, the components and the processes in the above embodiment may be combined with each

other. Modifications of the above embodiment will be described below. In the modifications described below, like reference numerals denote like structural elements in the above embodiment. Such elements will not be described.

(44) First Modification

(45) In a first modification, a worker corresponding to a detected face image is to be identified as a worker prestored in the storage **230** with the processing performed by the PC **200** described above. However, the worker identified based on image processing may be misidentified, causing the detected face region to be linked to a new worker ID. In this case, the above processing causes statistical information for a single worker to be divided into multiple sets of statistical information.

(46) In the example shown in FIG. 5, an image of the face region of the worker with a worker ID2 is misidentified as a new worker and receives another worker ID3. Thus, information for a single worker is calculated separately for the different worker ID2 and ID3, and different Gantt charts linked to the respective worker IDs are generated. As in the example of a display screen **500**, with the display screen **500** including a single worker linked to multiple Gantt charts as described above, a user can operate an input device (not shown) connected to the display and integrate sets of the face images and the Gantt charts to increase the efficiency of analyzing statistical information.

(47) More specifically, the user operates a cursor **501** displayed on the display screen **500** and selects two different face images **402a** and **402b** indicating the same worker. With the two face images **402a** and **402b** being selected, the user operates the input device and causes a menu **502** for integrating these two face images **402a** and **402b** to be displayed. The user further operates the input device and selects the menu **502** to perform integration of the two face images **402a** and **402b** selected by the PC **200** and the Gantt charts linked to the face images. Thus, the two sets of the face images **402a** and **402b** and the Gantt charts **407** and **408** selected on the display screen **500** are integrated into a single set of a face image and a new Gantt chart.

(48) The face image to be displayed on the display screen **500** after the integration may be one of the face images **402a** and **402b** selected by a user input or through the processing performed by the PC **200**, or another face image indicating the same worker stored in the storage **230**. The new Gantt chart may be generated by the statistical information generator **222** calculating the average of each process based on the original Gantt charts **407** and **408**. The user may operate the input device and, instead of selecting the two face images **402a** and **402b**, select the identification numbers (ID2 and ID3 in the figure) or the Gantt charts **407** and **408** displayed with a link to the face images **402a** and **402b**. Three or more sets of face images and Gantt charts may be combined.

(49) Second Modification

(50) In a second modification, the processing performed by the PC **200** described above generates data for a display screen **600** to appear on the display as illustrated in FIG. 6. In the present modification, as in the example of the display screen **600**, a menu **601** for allowing the PC **200** to automatically perform the integration of multiple sets of face images and Gantt charts as in the first modification appears on the display screen **600**. In the second modification as well, the background to a single worker linked to multiple sets of statistical information is the same as in the first modification. The second modification will be described focusing on the differences from the first modification.

(51) With the display screen **600** on the display, the user operates the input device and presses an execution button **602** in the menu **601** with the cursor **501**. This causes the PC **200** to perform the processing of integrating multiple sets of the face images **402a** and **402b** and the Gantt charts **407** and **408** as in the first modification. The PC **200** can identify the sets of the face images **402a** and **402b** and the Gantt charts **407** and **408** to be integrated based on the similarity between the face images.

(52) Third Modification

(53) In a third modification, the processing performed by the PC **200** described above generates data for a display screen **700** to appear on the display as illustrated in FIG. 7. In the present

modification, as in the example of the display screen **700**, the face of a worker displayed in a face image **702** is partially missing. The user operates the input device and selects a face image more appropriate as the face image **702** to facilitate distinguishing statistical information for each worker. (54) More specifically, the user operates the cursor **501** on the display screen **700** and selects the face image **702** to be changed. With the face image **702** being selected, the user operates the input device and causes the menu **701** for reselecting a face image **702** to be displayed. In response to the user further operating the input device and selecting the menu **701**, other face images of a worker corresponding to the face image **702** among face images generated by the image generator **221** and stored in the storage **230** appear on a selection screen **703**. In response to the user selecting an appropriate face image from the multiple face images displayed on the selection screen **703** and pressing a select button **704**, the face image **702** is changed to the selected face image. The resultant face image is linked to the worker ID and the statistical information and is stored into the storage **230** for subsequent face image displays.

(55) Fourth Modification

(56) In a fourth modification, the processing performed by the PC **200** described above generates data for a display screen **800** to appear on the display as illustrated in FIG. **8**. In the present modification, as in the example of the display screen **800**, in response to the user operating the input device and selecting one of face images (the face image **405** in the example in the figure) on the display screen **400** displayed in the first modification, more detailed information about the statistical information for operations performed by a worker corresponding to the selected face image **405** appears on the display screen **800**. Examples of information displayed on the display screen **800** include Gantt charts indicating the operation time per cycle on the cellular line (Gantt charts corresponding to C1 to C4 in the figure), a Gantt chart indicating the average operation time, and a Gantt chart indicating a target operation time. The details of the statistical information on the display screen **800** are not limited to the information illustrated in the figure and may be set as appropriate. This allows the user to switch the screen to view the outline information about the statistical information for operations associated with multiple workers (display screen **400**) and the detailed information about the statistical information for operations associated with an individual worker (display screen **800**) and to analyze the statistical information more efficiently.

(57) Fifth Modification

(58) In a fifth modification, the processing performed by the PC **200** described above generates data for a display screen **900** to appear on the display as illustrated in FIG. **9**. In the present modification, as in the example of the display screen **900**, the user can view a moving image of each operation performed by a worker.

(59) More specifically, the user operates the cursor **501** on the display screen **900** and selects, in a Gantt chart for a worker, a process to view its moving image (in the example in the figure, process 5 of a worker with the face image **402** is selected), and a view menu **901** for viewing a moving image appears. In response to the user operating the input device and selecting the view menu **901**, a moving image screen **902** for the selected process appears, and the moving image is played. Thus, the user can obtain information about the details of operations unobtainable from statistical information alone, and can perform more accurate analysis. For a moving image to be displayed on the moving image screen **902**, the controller **220** generates a moving image for each process by a worker based on the moving image data stored in the storage **230**, and may store the moving image into the storage **230**. Although a moving image is displayed for each process in the example described above, a moving image including a series of processes may be displayed to cover multiple processes or an entire set of processes.

(60) Sixth Modification

(61) In a sixth modification, the processing performed by the PC **200** described above generates data for a display screen **1000** to appear on the display as illustrated in FIG. **10**. In the present modification, as in the example of the display screen **1000**, the user can simultaneously view

moving images of multiple workers (or processes).

(62) More specifically, the user operates the cursor **501** on the display screen **1000** and selects, in the Gantt charts for workers, workers to view their moving images by checking checkboxes **1001** to **1005** appearing adjacent to the face images **401** to **405** of the workers (in the example in the figure, the worker with the face image **402** and the worker with the face image **404** are selected). In response to the user operating the input device and selecting a view menu **1006** for viewing moving images, moving image screens for each worker with a checked one of the checkboxes **1001** to **1005** (in the figure, a moving image screen **1012** corresponding to the worker **402** and a moving image screen **1014** corresponding to the worker **404**) appear, and the moving images are played. Thus, information unrecognizable from the statistical information may be obtained by comparing the behavior of a new worker with that of an expert worker during a process.

(63) Seventh Modification

(64) In a seventh modification, the processing performed by the PC **200** described above generates data for a display screen **1100** to appear on the display as illustrated in FIG. **11**. In the present modification, as in the example of the display screen **1100**, a notification area **1101** appears on the display screen **1100** to notify the user of a piece of statistical information for workers different from the target operation time by a threshold or greater. For example, as illustrated in the figure, the notification area **1101** stores the date of notification in its date column and the details of a notification in its details of notification column. In the example in the figure, the details of the notification, ID5 cycle 3 process 5 gap large (+6 minutes), indicates that the operation time of the worker with ID5 (the worker with the face image **405**) for the process 5 during the third cycle is 6 minutes longer than the target operation time. The information displayed in the notification area **1101** may be generated by the statistical information generator **222**. For example, the statistical information generator **222** compares statistical information for operations performed by each worker with the target operation time and, with the difference being a threshold or greater, generates information to be displayed in the notification area **1101**. Thus, the user uses the received information to efficiently perform more accurate analysis of the statistical information. The information to be displayed on the notification area **1101** is not limited to the example in the figure, and various items of information may be displayed. A criterion for determining whether to display information in the notification area **1101** may be determined as appropriate depending on the details of the information to be displayed.

(65) Eighth Modification

(66) In an eighth modification, the processing performed by the PC **200** described above generates data for a display screen **1200** to appear on the display as illustrated in FIG. **12**. The display screen **1200** shows periods (e.g., today, yesterday, last week, this week, and any other periods such as this month and last month) with different average values of the operation time for a worker, allowing comparison of operations performed by the same worker over the different periods. Thus, a change in the efficiency of the operations, for example, is viewed more easily. In the present modification, as in the example of the display screen **1200**, the Gantt charts that are statistical information for operations performed by a worker are compared with target values. The process with a shorter operation time than the target value by a predetermined threshold or greater is indicated as good (an icon **1202**, G, in the figure indicating good), and the process with a longer operation time than the target time by a predetermined threshold or greater is indicated as poor (an icon **1201**, NG, in the figure indicating not good). Evaluation information indicating the evaluation results of the statistical information may be information resulting from various evaluations. The evaluations may use predetermined criteria determined as appropriate depending on the evaluation. In response to the user operating the input device and selecting the process indicated as good or poor (the process with the icon **1201**, NG, is selected in the example in the figure) on the display screen **1200**, a moving image screen **1203** to show a frame-by-frame moving image of the selected process appears, and the moving image is played. The moving image to be displayed on the moving image

screen **1203** may be generated by the controller **220** for each operation performed by a worker based on the moving image data stored in the storage **230**, and may be stored in the storage **230**.

(67) The embodiments and modifications described above may be combined as appropriate. In the third modification, a face image of a worker is changed based on a user input. However, the PC **200** may automatically change the image to, for example, an image with the face region appearing larger in size, an image with the face region facing more to the front based on estimation of the orientation of the face, or an image more reliable as the face region, based on the face images stored in the storage **230**. In some embodiments, the storage **230** prestores the face images of workers (e.g., the face images used for employee cards), worker IDs, and names as worker information, and the face images of workers to be displayed on the display screens described in the above embodiments and modifications may be changed to the worker information stored in the storage **230**. In this case, the data generator **223** performs matching using the face images of workers generated by the image generator **221** and the face images of workers stored in the storage **230** based on the degree of similarity in features between the images. The data generator **223** replaces, based on the matching result, the face image to be displayed on the display screen described above with a face image with a high degree of similarity among the face images of workers stored in the storage **230**, and generates data for the display screen by linking the replaced face image to the Gantt chart of the worker. Through this processing, the user can more efficiently distinguish each worker and analyze statistical information.

(68) In the fourth modification, the details of statistical information for operations associated with individual workers are displayed on the display screen **800**. In addition to or instead of the information, the details of the operation time per process, the evaluation results for each process, or information indicating results based on analysis results may be displayed. The evaluation or analysis to be used may be determined as appropriate.

(69) Appendix 1

(70) An image processing apparatus, comprising: an image obtainer (**210**) configured to obtain a moving image including an operation area captured with an image sensor; an image generator (**221**) configured to generate an image of a face region of a worker performing an operation in the operation area based on the moving image; a statistical information generator (**222**) configured to generate statistical information for the operation performed by the worker based on the moving image; a data generator (**223**) configured to generate data including the generated image of the face of the worker linked to the generated statistical information for the operation performed by the worker; and an output unit (**240**) configured to output the generated data.

Appendix 2

(71) An image processing method, comprising: (S**301**) obtaining a moving image including an operation area captured with an image sensor; (S**305**) generating an image of a face region of a worker performing an operation in the operation area based on the moving image; (S**307**) generating statistical information for the operation performed by the worker based on the moving image; (S**308**) generating data including the generated image of the face region of the worker linked to the generated statistical information for the operation performed by the worker; and (S**309**) outputting the generated data.

Claims

1. An image processing apparatus, comprising: a memory configured to store computer-executable instructions; and a processor configured to execute the computer-executable instructions stored in the memory to implement; an image obtainer configured to obtain a moving image including an operation area captured with an image sensor; an image generator configured to generate an image of a face region of a worker performing an operation in the operation area based on the obtained moving image; a statistical information generator configured to generate statistical information for

the operation performed by the worker for each of one or more operational processes included in the operation area based on the obtained moving image; a data generator configured to generate data including the generated image of the face of the worker linked to the generated statistical information for the operation performed by the worker; and an output unit configured to output the generated data, wherein the data generator generates, based on the obtained moving image, a moving image associated with the generated statistical information for the operation performed by the worker for each of the one or more operational processes, the output unit outputs the generated moving image for each of the one or more operational processes to a display, and the generated moving image for each of the one or more operational processes is displayed on a display screen of the display.

2. The image processing apparatus according to claim 1, wherein the data generator generates data including images of face regions of a plurality of workers generated by the image generator linked to statistical information for operations performed by the plurality of workers generated by the statistical information generator, and the data generator generates, for data output from the output unit, new data including at least two images of the images of the face regions of the plurality of workers integrated with the statistical information linked to the at least two images.

3. The image processing apparatus according to claim 2, wherein the at least two images are selected based on a user input.

4. The image processing apparatus according to claim 1, wherein the data generator generates, for data output from the output unit, new data including the image of the face region of the worker replaced by another image of the face region of the worker generated by the image generator.

5. The image processing apparatus according to claim 1, wherein the data generator generates data including an image of the face region of the worker selected based on a criterion including at least one of a size, an orientation, or a facial feature of the face region of the worker linked to the generated statistical information for the operation performed by the worker.

6. The image processing apparatus according to claim 1, wherein the data generator generates, based on matching between images of face regions of a plurality of workers obtained in advance and the generated image of the face region of the worker, data including an image of the face region of the worker selected from the images obtained in advance linked to the generated statistical information for the operation performed by the worker.

7. The image processing apparatus according to claim 1, wherein the statistical information generator generates outline information and detailed information about the statistical information for the operation performed by the worker based on the moving image, and the data generator generates data including the image of the face region of the worker linked to the outline information about the generated statistical information for the operation performed by the worker and the detailed information about the generated statistical information for the operation performed by the worker.

8. The image processing apparatus according to claim 1, wherein the data generator generates a moving image associated with the generated statistical information for the operation performed by the worker based on the obtained moving image, and the data generator generates data including the image of the face region of the worker linked to the generated statistical information for the operation performed by the worker and the generated moving image.

9. The image processing apparatus according to claim 1, wherein the data generator generates data including the image of the face region of the worker linked to the generated statistical information for the operation performed by the worker for each of the one or more operational processes and the generated moving image for each of the one or more operational processes.

10. The image processing apparatus according to claim 1, wherein the statistical information generator generates evaluation information indicating an evaluation result obtained based on a predetermined criterion for the statistical information for the operation performed by the worker generated based on the moving image, and the data generator generates data including the image of

the face region of the worker linked to the generated statistical information for the operation performed by the worker and the evaluation information.

11. An image processing method, comprising: obtaining a moving image including an operation area captured with an image sensor; generating an image of a face region of a worker performing an operation in the operation area based on the obtained moving image; generating statistical information for the operation performed by the worker for each of one or more operational processes included in the operation area based on the obtained moving image; generating data including the generated image of the face region of the worker linked to the generated statistical information for the operation performed by the worker; outputting the generated data; generating, based on the obtained moving image, a moving image associated with the generated statistical information for the operation performed by the worker for each of the one or more operational processes; and outputting the generated moving image for each of the one or more operational processes to a display, wherein the generated moving image for each of the one or more operational processes is displayed on a display screen of the display.

12. A non-transitory computer readable medium storing a program for causing a computer to perform a process, comprising: obtaining a moving image including an operation area captured with an image sensor; generating an image of a face region of a worker performing an operation in the operation area based on the obtained moving image; generating statistical information for the operation performed by the worker for each of one or more operational processes included in the operation area based on the obtained moving image; generating data including the generated image of the face region of the worker linked to the generated statistical information for the operation performed by the worker; outputting the generated data; generating, based on the obtained moving image, a moving image associated with the generated statistical information for the operation performed by the worker for each of the one or more operational processes; and outputting the generated moving image for each of the one or more operational processes to a display, wherein the generated moving image for each of the one or more operational processes is displayed on a display screen of the display.
